SECURITY EVENT CLASSIFICATION

Practical Report | Cybersecurity Intern Task

Task	Classify sample security events as Normal, Suspicious, or Potentially Malicious
Tools	Microsoft Excel / spreadsheet
Objective	Develop basic security-triage skills by reviewing events and documenting why each event received its classification.

1. Overview

This practical involved examining a sample set of ten security events and assigning a category to each event. A short reason was recorded alongside every classification. This is a basic form of **security triage**: reviewing alerts or activity to decide what appears routine and what may need further investigation.

2. Method Followed

• Reviewed each event description in the spreadsheet.

• Selected a category based on the described behaviour: Normal, Suspicious, or Potentially Malicious.

• Recorded a reason explaining the decision, such as expected user activity, unusual location, repeated failed logins, or possible data theft.

• Counted the number of events in each category and wrote a short conclusion.

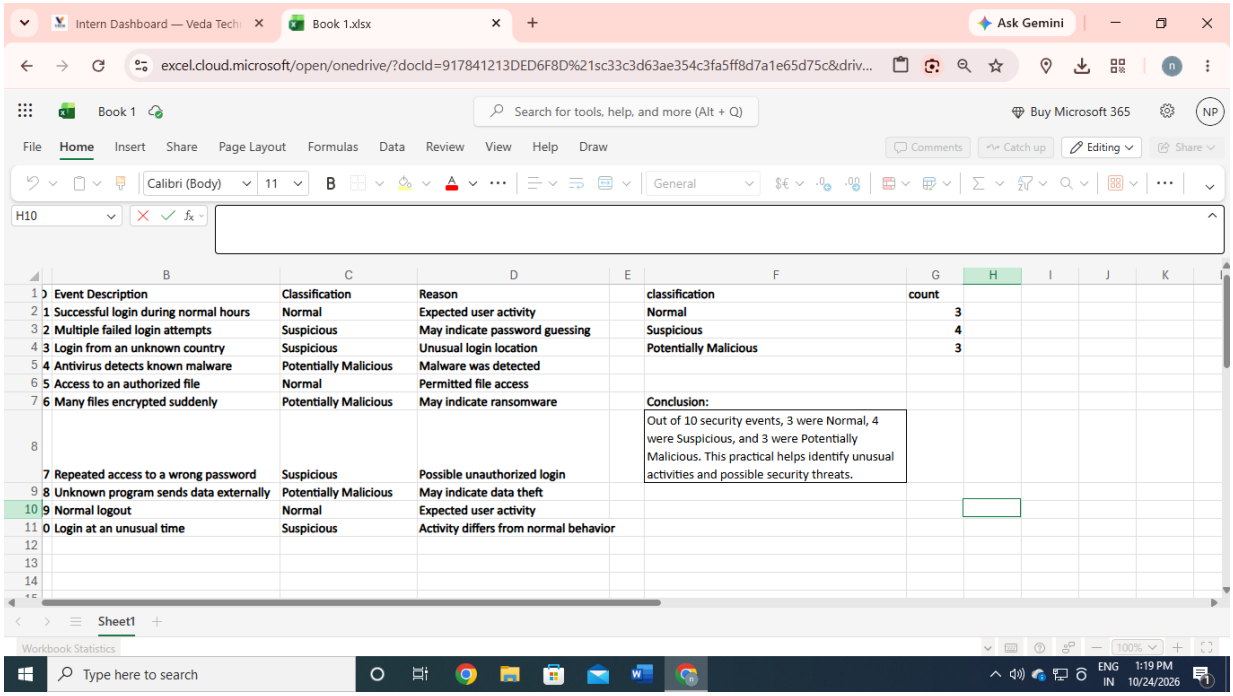
3. Results Summary

Classification	Count	Meaning in this exercise
Normal	3	Activity that appears consistent with expected use
Suspicious	4	Unusual activity that may require checking
Potentially Malicious	3	Activity that could indicate a threat and needs investigation
Total	**10**	**All sample events classified**

Result: 30% of the sample events were Normal, 40% were Suspicious, and 30% were Potentially Malicious. These labels are based only on the sample descriptions; they do not prove that an attack actually occurred.

4. Spreadsheet Evidence

The screenshot below shows the completed spreadsheet, including event descriptions, classifications, reasons, category counts, and the written conclusion. The separate task-instruction screenshot was used only as reference for preparing this report and is intentionally not attached.



	B	C	D	E	F	G	H	I	J	K
1	Event Description	Classification	Reason		classification	count				
2	1 Successful login during normal hours	Normal	Expected user activity		Normal	3				
3	2 Multiple failed login attempts	Suspicious	May indicate password guessing		Suspicious	4				
4	3 Login from an unknown country	Suspicious	Unusual login location		Potentially Malicious	3				
5	4 Antivirus detects known malware	Potentially Malicious	Malware was detected							
6	5 Access to an authorized file	Normal	Permitted file access							
7	6 Many files encrypted suddenly	Potentially Malicious	May indicate ransomware		Conclusion:					
8	7 Repeated access to a wrong password	Suspicious	Possible unauthorized login		Out of 10 security events, 3 were Normal, 4 were Suspicious, and 3 were Potentially Malicious. This practical helps identify unusual activities and possible security threats.					
9	8 Unknown program sends data externally	Potentially Malicious	May indicate data theft							
10	9 Normal logout	Normal	Expected user activity							
11	0 Login at an unusual time	Suspicious	Activity differs from normal behavior							
12										
13										
14										

Figure 1. Excel worksheet showing the classified security-event dataset.

5. Key Learning

• Security triage helps organise events so analysts can decide which ones need attention first.

• A classification should include a clear reason, not just a label.

• Repeated failed logins, logins from unusual countries, unexpected external data transfers, and sudden file encryption can be warning signs.

• A suspicious event should be investigated using additional logs and context before drawing a final conclusion.

6. Conclusion

The task was completed by classifying ten sample security events and documenting the reason for each decision. The spreadsheet summarises the results as 3 Normal, 4 Suspicious, and 3 Potentially Malicious events. This exercise demonstrates the basics of security-event review and the importance of evidence-based classification. In a real environment, the next step would be to verify suspicious activity against additional logs and escalate confirmed or high-risk findings according to the organisation's incident-response process.

Note: This report describes a training exercise using sample/synthetic event descriptions. It does not claim that real systems were compromised or that a real security incident was confirmed.